

B3 Uniform Accelerated Motion in One Dimension

Assume that any dropped or thrown object accelerates downwards at 9.81 m s^{-2} . If a question says that an object is 'dropped', this means that its velocity is zero at the beginning of the motion.

If asked for a velocity or displacement, your answer **MUST** contain a direction in order to be marked as correct.

- B3.1 A calculator falls off a desk which is 0.90 m high. How much time passes before it hits the floor?
- B3.2 A potato is fired vertically into the air from the ground at a speed of 45 m s^{-1} .
- a) How high up is it when it is travelling at 15 m s^{-1} upwards?
 - b) How high up is it when it is travelling at 15 m s^{-1} downwards?
- B3.3 The Highway Code states that the stopping distance for a car travelling at 31 m s^{-1} is 96 m . If the data assumes that the reaction time of the driver is 0.67 s , calculate the deceleration of the car once the brakes have been applied.

CHAPTER B. MECHANICS

Answers

B3.1 $\sqrt{\left(\frac{2 \times 0.9}{9.81}\right)} = 0.428 \text{ s}$

B3.2

a) $\frac{45^2 - 15^2}{2 \times 9.81} = 91.74 \text{ m s}^{-1}$

b) $\frac{45^2 - 15^2}{2 \times 9.81} = 91.74 \text{ m s}^{-1}$ as in part (a)

B3.3 $\frac{-31^2}{2(96 - 31 \times 0.67)} = -6.387 \text{ m s}^{-2}$
So deceleration is 6.387 m s^{-2}